

MA 511: Real Analysis

Course Code	MA 511
Course Name	Real Analysis
L-T-P-C	3-1-0-4
Prerequisites	None
Intended for	M.Sc. /M.S./Ph.D. /B.Tech 3rd and 4th year
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline.
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I** [4 Lectures]
Introduction to real numbers, Construction, Dedekind cuts, Completeness property, Archimedean property, Countable and uncountable set
- **Module II** [6 Lectures]
Open balls and open sets in Euclidean space, Definition of interior points, Closed sets, Adherent points, Accumulation points, Closure, Bolzano-Weirstrass Theorem, Cantor intersection theorem, Heine-Borel Theorem, Compactness.
- **Module III** [7 Lectures]
3. Metric spaces, Open sets, Closed sets, Dense sets, Metric subspaces, Compact subsets of a metric space, Boundary of a set, Totally boundedness, Completeness.
- **Module IV** [8 Lectures]
Convergent sequences in a metric space, Cauchy sequences, Complete metric space, Limit of a function, Continuous functions, Continuity of composite functions, Continuity and inverse image of open and closed sets, Functions continuous on compact sets, Connectedness
- **Module V** [4 Lectures]
Review of Riemann Integration, Riemann-Stieltjes integral: definition and examples, Properties of the integral.
- **Module VI** [10 Lectures]
6. Uniform continuity, Fixed point theorem for contractions, Sequences of functions, Point wise convergence of sequences of functions, Uniform convergence of sequences of functions, Uniform convergence and continuity, Cauchy condition for uniform convergence, Uniform convergence of infinite series of functions, Cauchy condition for uniform convergence of series, Weirstrass M-test, Dirichlet's test for uniform convergence, Uniform convergence and differentiation, Uniform convergence and integration
- **Module VII** [Lectures]
Metric space $C[a,b]$, Characterize compact subsets, i.e., Arzela-Ascoli theorem.

Text Books

1. W. Rudin, **Principles of Mathematical Analysis**, 3rd Edition, McGraw-Hill, 2013.
2. T. Apostol, **Mathematical Analysis**, 2nd Edition, Narosa Publishers, 2002.

References

1. Elias M. Stein and Rami Shakarchi, **Real Analysis** Princeton Lectures, 2010.
2. Terrance Tao, **Analysis I and II**, Trim, Hindustan book agency, 2006.